Reg. No										

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CH102T – KINETIC THEORY OF GASES AND CHEMICAL KINETICS

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. 1 atm= ---------
   a) 720 mm Hg b) 760 mm Hg c) 740 Torr d) 720 Torr

2. V∞T (n,P are constant) is ---------- law
   a) Boyle's b) Charle's c) Molefraction d) Avogadro's

3. The temperature at which the second virial coefficient of a real gas is zero is called
   a) Critical temperature b) Eutectic point
   c) Boiling point d) Boyle temperature

4. Intermolecular attraction and molecular size of a gas can be neglected at
   a) Tc b) High pressure
   c) Low P and high T d) High P and low T

5. A ------order reaction is one whose rate is independent of concentration.
   a)1/2 b) 1 c) 0 d) 2

6. In ---------catalysis, the catalyst is in the same phase as the reactants.
   a) Homogeneous b) Heterogeneous
   c) Enzyme reaction d) All

7. ------- can act as enzyme inhibitor
   a) Ag+ b) Br- c) Cl- d) H+

8. Promoter is not a catalyst but ---------- the activity of it.
   a) Decreases b) Increases c) Enzymes d) Poisons

9. --------- adsorption is due to van der Waals attractive forces.
   a) Chemical b) Solvent c) Physical d) Solution

10. Freundlich isotherm is not applicable at high----------
    a) Pressure b) Temperature c) Speed d) Concentration

II. Fill up the blanks

11. ---------- is the formula used to find RMS velocity.

12. ---------- is the concept behind Linde's refrigerator.

13. The number of reactant molecules involved in a reaction is called ----------.

14. A---------- in the surface area increases the extent of adsorption.

15. The rate of enzyme catalyzed reactions is maximum at the ---------- temperature.

III. Match the following

16. Newton Meter - Joule Thomson effect

17. Isenthalpic process - $kgms^{-2}$

18. Partial pressure - $mol^{-1}dm^3s^{-1}$

19. Second order - Optimum pH

20. 7.4 - Molar concentration X RT

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. What is most probable velocity?

22. What is Graham's law of diffusion?

23. What is compressibility factor?

24. Write a note on collision diameter.

25. What is the effect of molar mass on molecular velocities?

26. Calculate Kp for the reaction $O_2(g) \leftrightarrow O_3(g)$ at 298K.ΔG=163.43kJmol^{-1}.

27. What are the applications of Adsorption?

28. Calculate the units of the rate constants for i) Zero, ii) 1storder.

29. Differentiate Order and Molecularity.

30. What is Activated complex theory?

31. What is chain reaction?

32. Write a note on explosion limit.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a. Derive ideal gas equation. **(4)**
    b. Convert the following into ^{o}C a)-32K, b) boiling point of water in ^{o}F. **(3)**

34. a. Write a note on effusion? **(3)**
    b. Derive the relationship between Average and RMS velocity. **(4)**

35. a. Derive the relationship between Kp&Kc. **(4)**
    b. Explain Homogeneous catalysis. **(3)**

36. a. Explain Enzyme inhibition. **(3)**
    b. Define the terms i) adsorption, ii) adsorbate. **(4)**

37. a. Explain Pseudo order reaction with an example. **(4)**
    b. Explain the significance of BET theory. **(3)**

38. a. For a 1st order reaction t½ is 100 seconds. How long will it take for the reaction to get completed 75%? **(4)**
    b. Explain temperature coefficient. **(3)**

39. a. Derive Michaelis- Menton equation. **(7)**

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